Understanding Robot Pose

Lesson completed | Isaac Sim 6.1 | 2D and introductory 3D pose

Learning objectives

Define pose as position and orientation; represent planar and introductory 3D pose; reason about fixed and rotated reference frames; edit a cube's pose in Isaac Sim.

Concept 1 | What is pose?

Pose includes both position and orientation, expressed relative to a reference frame. A robot changing from (4, 2, 90 degrees) to (4, 2, 180 degrees) changes pose despite staying at the same position.

Concept 2 | Pose representation

Planar pose: P = (x, y, theta). Example: (2, 3, 0 degrees) plus world-frame motion (+2, +1) and rotation +90 degrees becomes (4, 4, 90 degrees). Use theta_final = theta_initial + signed rotation. Introductory 3D shorthand: (x, y, z, roll, pitch, yaw); Euler rotation order and convention must be defined.

Concept 3 | Frames of reference

Robot at world (3, 2), cup at world (7, 5): displacement (4, 3). If robot yaw = 0 degrees, cup relative to robot = (4, 3). After robot yaw = +90 degrees, the robot-frame coordinates = (3, -4), while both world positions remain unchanged.

Understanding correction

The learner initially expected robot-frame coordinates to stay constant when only the robot turned. After explanation, they correctly identified that a rotated coordinate frame produces different coordinate values.

Concept 4 | Isaac Sim 6.1

The learner reported setting a cube Transform to first Translate (2, 1, 1.5), Rotate (0, 0, 45 degrees), then second Translate (4, 1, 1.5), Rotate (0, 0, 90 degrees). Correct changes: +2 in Translate X and +45 degrees in Rotate Z. The learner confirmed the scene was saved as lesson_04_robot_pose.usd.

Evidence and limitations

The Isaac Sim scene file was not uploaded; the simulator transform cannot be independently inspected here. Code checks independently verify the arithmetic, not the scene. The saved USD generally stores the final state, not the intermediate sequence. The values are world-frame poses only under the stated no-transformed-parent assumption.

Learning review

1. Does pose change when orientation changes but position stays fixed? Yes. 2. What is the planar pose after motion from (2, 3, 0 degrees)? (4, 4, 90 degrees). 3. Cup relative to robot at yaw 0? (4, 3). 4. After robot yaw +90? (3, -4). 5. Isaac Sim changes? Translate X +2; Rotate Z +45 degrees.

Next steps

Keep the original USD file alongside the evidence package. Rotation matrices, quaternions, composing transforms and ROS TF remain future subjects. Lesson 5 is not started.